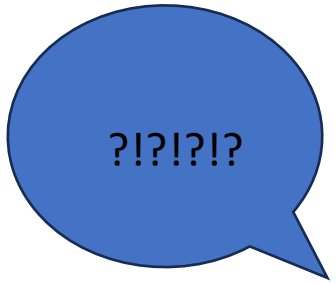


Uncertainty

Xuanji Yu
Courtesy of Professor Roch

12.2-12.6,
13-13.3.3.1
(except 13.2.1),
Rabiner's HMM Tutorial

Uncertainty

- The real world is rife with uncertainty!
  - E.g., if I leave for SAN 60 minutes before my flight, will I be there in time?

- Problems:
  - partial observability (road state, other drivers' plans, etc.)
  - noisy sensors (radio traffic reports, Google maps)
  - immense complexity of modelling and predicting traffic, security line, etc.
  - lack of knowledge of world dynamics (will tire burst? need COVID test?)

Uncertainty

$$toothache \Rightarrow cavity$$

What else can cause a toothache?

$$toothache \Rightarrow cavity \vee cracked\ tooth \vee stuck\ popcorn \vee \ldots$$

Logic can fail us:

- laziness – Too difficult to enumerate rules without exceptions
- theoretical ignorance – May not fully understand the system
- practical ignorance – System may not be fully observable

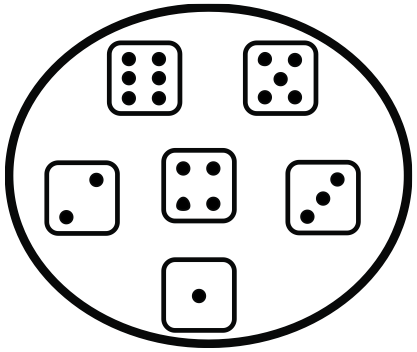
An agent’s view

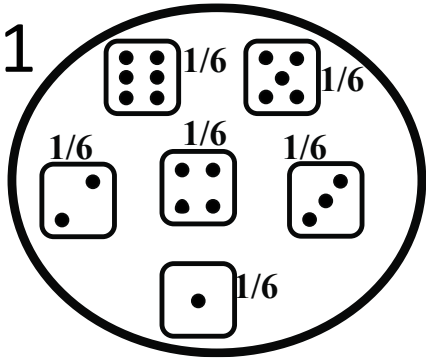
Probabilities represent a level of belief in a world

Decision-theoretic agents

```
function DT-AGENT(percept) returns an action
  persistent: belief_state, probabilistic beliefs about the current state of the world
              action, the agent's action

  update belief_state based on action and percept
  calculate outcome probabilities for actions,
      given action descriptions and current belief_state
  select action with highest expected utility
      given probabilities of outcomes and utility information
  return action
```

Basic probability





- Random variables represent an outcome or world, e.g. X represents a die roll, $W_{2,1}$ is the Wumpus in cave 2,1
- P(X) is the probability of X happening
- It is common to use
  - CAPITALS to represent outcomes in general: P(X)
  - lower case to denote specific outcomes: P(x=5)
- Probability distributions characterize probability over all outcomes and require:

$$\forall x \in \text{domain}(X), 0 \leq P(X = x) \leq 1$$

$$\sum_{x \in \text{domain}(X)} P(X = x) = 1$$

Posterior (conditional) probability

- Posterior probability is conditioned on another event

  What is the probability that I have a cavity *given* that I have a toothache?

$$P(cavity|toothache)$$

- In contrast, prior (unconditional) probabilities have no condition, refering to degrees of belief in propositions in the absence of any other information

$$P(cavity)$$

- Definition: $\mathrm{P(A|B)} = \frac{P(A \wedge B)}{P(B)}$ or equivalently: $P(A \wedge B)$=P(A|B)P(B)

note: $P(A \wedge B)$ is frequently written as $P(A, B)$

product rule

Propositions

- Let us consider random variable values as possible worlds (like our model checking in propositional logic)
- If we want to know P of proposition ϕ holding:

$$P(\phi) = \sum_{\omega \in \phi} P(\omega)$$

- In addition

$$P(\neg\phi) = \sum_{\omega \in \neg\phi} P(\omega) = 1 - \sum_{\omega \in \phi} P(\omega)$$

and

$$P(\phi \vee \rho) = P(\phi) + P(\rho) - P(\phi \wedge \rho)$$

inclusion-exclusion principle

Joint probabilities

- Probability of multiple things, e.g. $P(A, B, C)$.
- Can be decomposed with the product rule:

$$\begin{aligned} P(A, B, C) &= P((A, B), C) \\ &= P(C|A, B)P(A, B) \\ &= P(C|A, B)P(B|A)P(A) \end{aligned}$$

  This is called the chain rule.
  In general: $P(x_1, \dots, x_n) = \Pi_{i=1}^{n} P(x_i | x_{i-1}, x_{i-2}, \dots, x_1)$
- If A and B are independent, then $P(B|A) = P(B)$
- In general, joint probabilities of independent variables can be multiplied: $P(A, B, C) = P(A)P(B)P(C)$ (A/B/C independent)

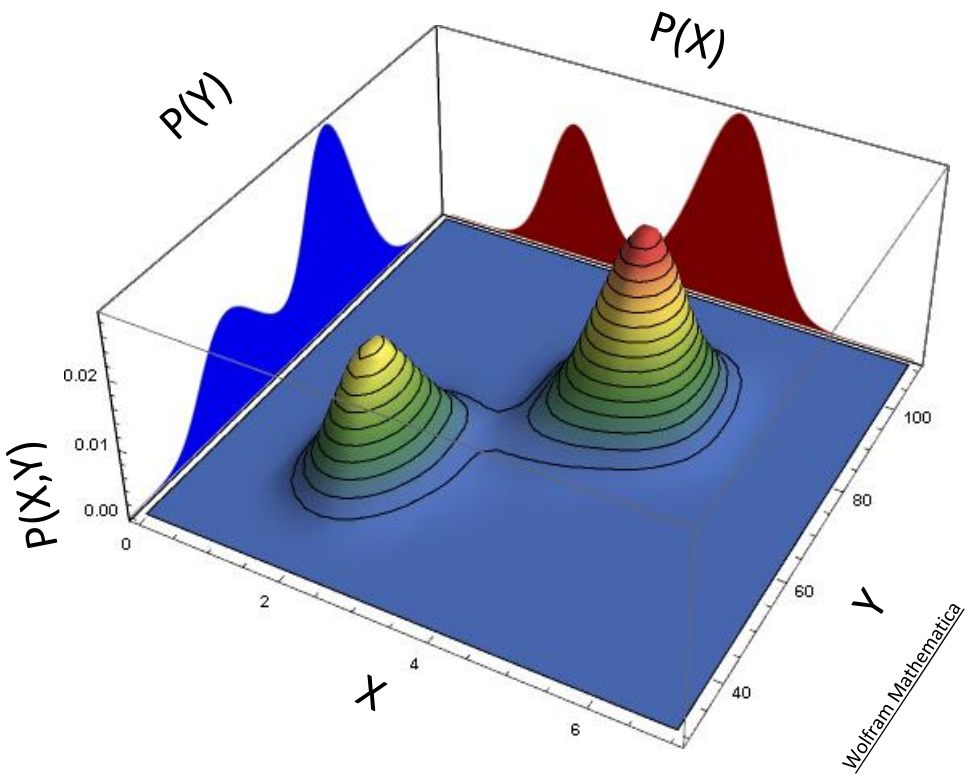
Marginalization

- Suppose we know the joint probability between X and Y, P(X,Y), and want P(X)

$$P(X) = \sum_y P(X, Y = y)$$

example:

$$P(Eat, Rent) = \sum_{r=0}^{owed} P(Eat, Rent = r)$$



- Marginaliztion: or called summing out, because we sum up the probability for each possible value of other variables, thereby taking them out of the equation

Bayes' rule

(of conditional probability)

Remember definition posterior probability $P(A|B) = \frac{P(A,B)}{P(B)}$

$$\mathrm{P(B|A)} = \frac{P(B,A)}{P(A)} \rightarrow P(B,A) = P(B|A)P(A)$$

$$P(A|B) = \frac{P(A,B)}{P(B)} = \frac{P(B,A)}{P(B)}$$

$$\therefore P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

also known as Bayes' Theorem/Bayes' Law)

This simple equation underlies most modern AI systems for probabilistic inference.

Bayes' rule: Why do we care?

Suppose we observe an effect.

- Knowing the cause can be difficult
- Simpler to estimate P(effect | cause); Bayes' rule lets us turn this around:

$$P(Disease|Symptom) = \frac{P(Symptom|Disease)P(Disease)}{P(Symptom)}$$

If we are looking at multiple diseases, we do not need P(Symptom) to make a choice between them.

We can treat ${}^{1}/_{P(Symptom)}$ as a constant α:

$$P(Disease|Symptom) = \alpha P(Symptom|Disease)P(Disease)$$

Bayes' rule example

- A symptom of meningitis is a stiff neck

$$P(s|m) = 0.7$$

but the case rate for stiff necks is low and meningitis very low

$$P(s) = 0.01, P(m) = 1/50000$$

$$P(m|s) = \frac{P(S|m)P(m)}{P(s)} = \frac{0.7 \times {}^{1}/_{50000}}{0.01} = 0.0014$$

Conditional independence

What do you see?
What does your neighbor see?

Conditional independence

- Both Shyam and Monica observe the cloud
- If they haven't talked to each other about what they saw, the probability conditioned on a specific cloud is independent

$$P(S, M|c) = P(S|c)P(M|c)$$

- Separation

$$P(S, M, c) = P(S, M|c)\, P(c)$$
$$= P(S|c)P(M|c)\, P(c)$$

Decomposition of large probabilistic domains into weakly connected subsets through conditional independence

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Naïve Bayes models

- Exploit conditional independence to make simple models
- If a cause has *n* effects that are conditionally independent:

$$P(cause, e_1, e_2, \dots, e_n) = P(cause) \prod_{i=1}^{n} P(e_i|cause)$$

- Naïve Bayes models imply that we don't really know if our effects are conditionally independent, but we assume so anyway.
- If we wanted to find $P(cause|e_1, e_2, \dots, e_n)$ we could use the product rule and conditional independence:

$$P(cause|e_1, e_2, \dots, e_n) = \alpha P(cause) \prod_{i=1}^{n} P(e_i|cause)$$

α is *normalization constant*

Interpreting equation: for each possible cause, multiply the prior probability of the cause by the product of the conditional probabilities of the observed effects given the cause, then normalize the result

Example: Sentence to category

Disneyland raised its entrance price by thirty percent.

We might ask the question: Is this about business or entertainment?

We could consider how often articles are about each of these categories (prior or unconditional):

- P(business) = .03, Prior probability of business change
- P(entertainment) = .04, Prior probability of entertainment improvements

Example: Sentence to category

Bayes factors:

In a scenario where multiple theme parks exist, each with varying business models and pricing strategies, we might observe a business model change or entertainment improvement at some random theme park and want to know if it’s likely to be Disneyland.

- P(Disneyland|business) = .2,
- P(Disneyland|entertainment) = .8,

Example: Sentence to category

Bayes factors:

Probability of price increase for different motives

- P(price|business) = .9, Probability of price increase if motive is business
- P(price|entertainment) = .1, Probability of price increase if motive is entertainment

Example: Sentence to category

$$P(cause|e_1, e_2, \dots, e_n) = \alpha P(cause) \prod_{i=1}^{n} P(e_i|cause)$$

Bayes factors:

P(Disneyland|business) = .2

P(Disneyland|entertainment) = .8

P(price|business) = .9

P(price|entertainment) = .1

Prior probabilities:

P(business) = .03

P(entertainment) = .04

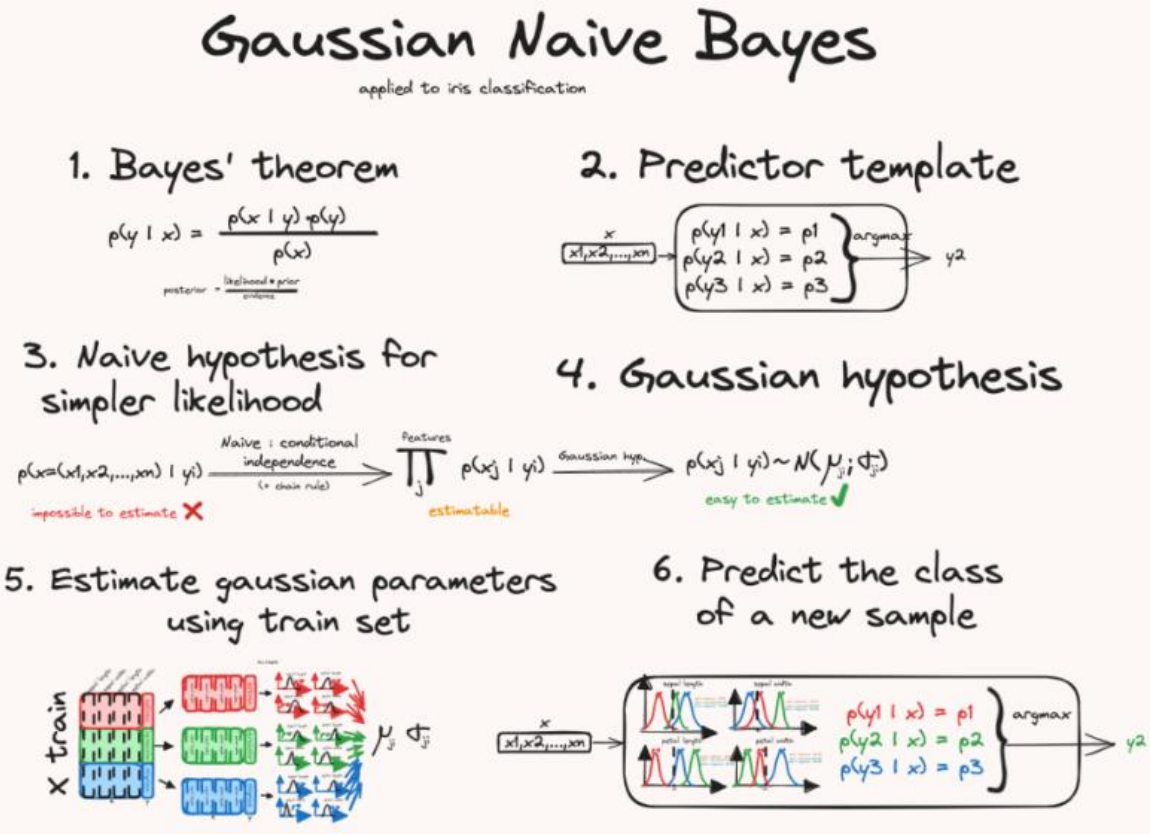
$$P(bz|Dland, \$)$$
$$= P(bz)P(Dland|bz)P(\$|bz)$$
$$= .03 \cdot .2 \cdot .9 = .0054$$

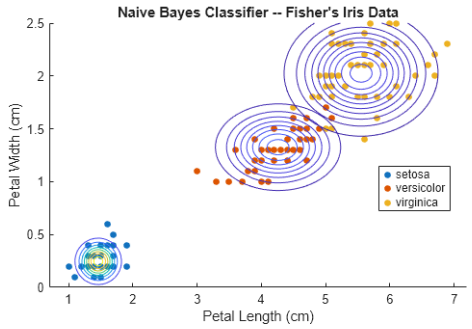
$$P(ent|Dland, \$)$$
$$= P(ent)P(Dland|ent)P(\$|ent)$$
$$= .04 \cdot .8 \cdot .1 = .0032$$

We classify as the sentence as the category that maximizes P

Example: Gaussian Naïve Bayes Classifier

```python
# Import libraries
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load the Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Initialize the Gaussian Naive Bayes model
gnb = GaussianNB()

# Train the model
gnb.fit(X_train, y_train)

# Make predictions
y_pred = gnb.predict(X_test)

# Evaluate performance
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
```





Probabilistic reasoning

Build efficient network models to reason under uncertainty according to the laws of probability theory, dinstinguish between correlation and causality

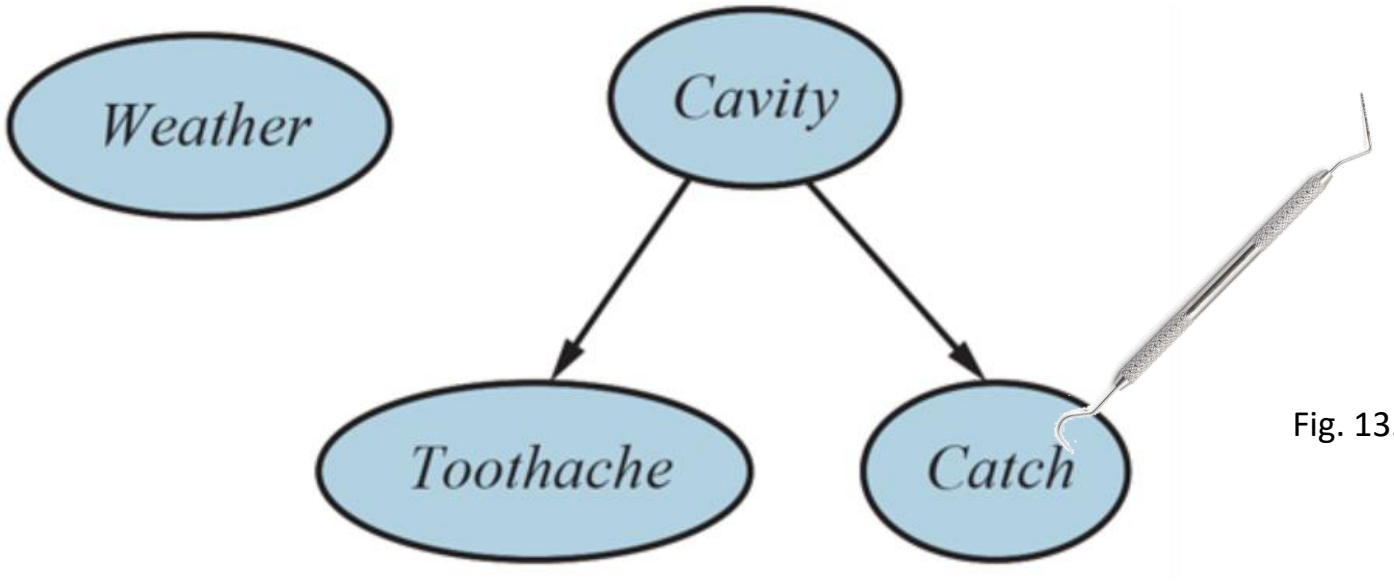


Fig. 13.1

A simple Bayesian network in which Weather is independent of the other three variables and Toothache and Catch are conditionally independent, given Cavity.

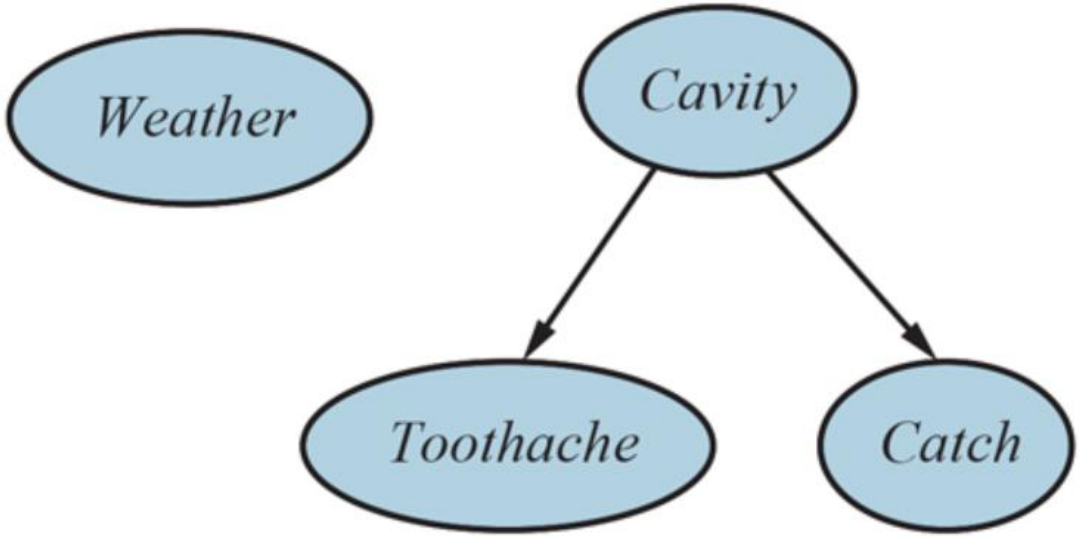
Bayesian network (represent dependencies among variables)

- Having a cavity influences the likelihood of a toothache or a dentist's sickle probe to catch on your tooth
- Changes in weather do not cause toothaches or probe catches.

Bayesian networks

- Nodes are random variables
- Variables can be connected by directed arcs that do not form cycles
- Each variable *V* has
  - prior probability (no parents): P(V)
  - conditional probability P(V|parents(V))
- Forms a directed acyclic graph

A simple Bayesian network in which Weather is independent of the other three variables and Toothache and Catch are conditionally independent, given Cavity.



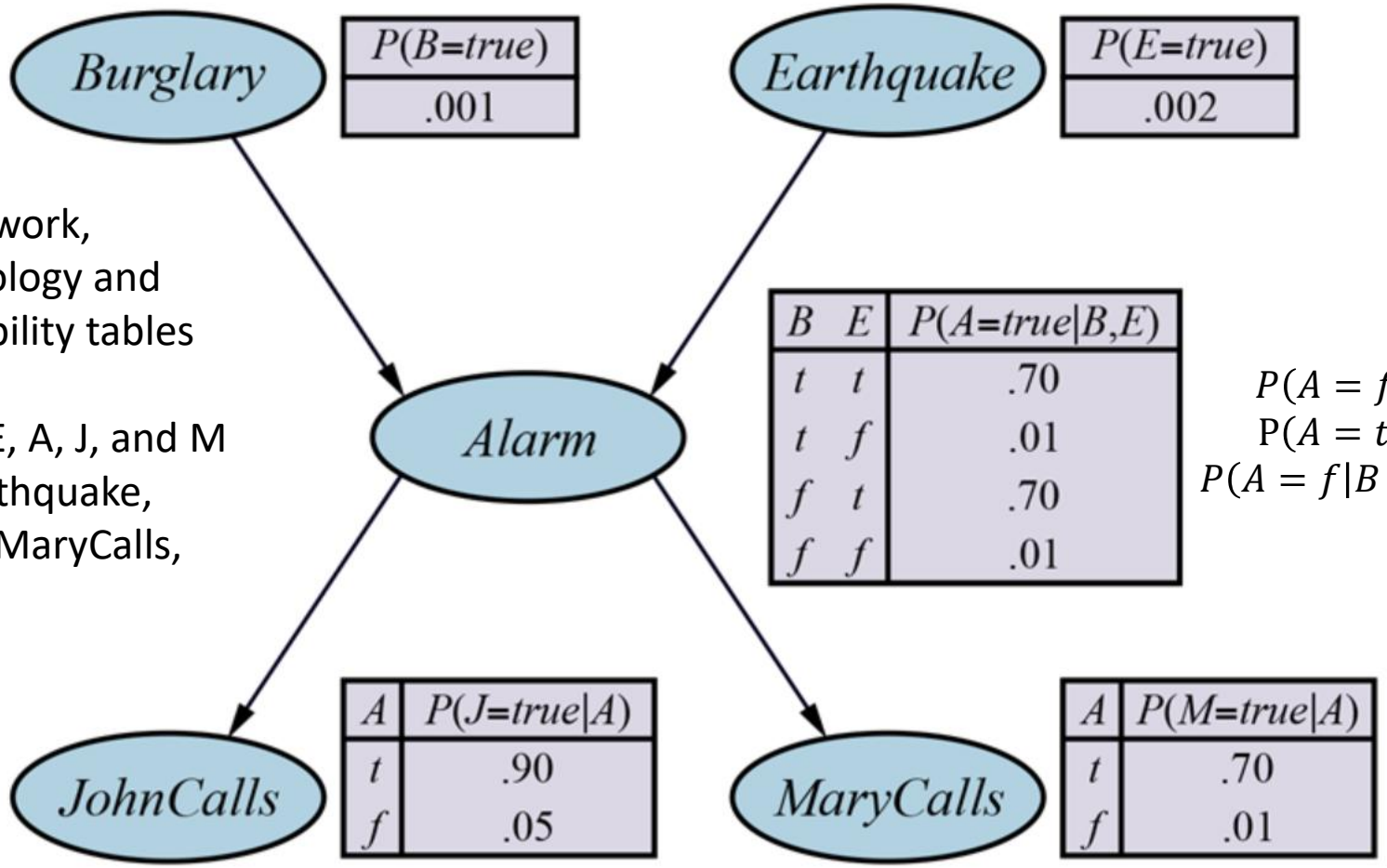
Pearl's Bayesian network example

You have a new burglar alarm installed at home.

- Burglar alarm set off by
  - Burglar
  - Occasionally Earthquake
- Neighbors Mary and John have agreed to call you at work when they hear the alarm
  - Mary listens to loud music, and often misses the alarm
  - Your home telephone ringtone is similar to the alarm (silly you) and John sometimes calls you when your phone rings
    (yes, you still have a landline)
- Given the evidence of who has or has not called, estimate the probability of a burglary

Pearl's Bayesian network example

- A typical Bayesian network, showing both the topology and the conditional probability tables (CPTs).
- In the CPTs, letters B, E, A, J, and M stand for Burglary, Earthquake, Alarm, JohnCalls, and MaryCalls, respectively



$P(A = f|conditions)$ implicit
$\mathrm{P}(A = t|B = t, E = t) = .7 \rightarrow$
$P(A = f|B = t, E = t) = 1 - .7 = .3$

Fig. 13.2 R&N

Bayes net semantics

$P(A = f|conditions)$ implicit
$\mathrm{P}(A = t|B = t, E = t) = .7 \rightarrow$
$P(A = f|B = t, E = t) = 1 - .7 = .3$

- $P(x_1, x_2, \dots, x_n) = \prod_{i=1}^{n} P(x_i|parents(X_i))$
  - $parents(X_i)$ denotes values of $parents(X_i)$ that appear in $x_1, x_2, \dots, x_n$
  - Each entry in the join distribution is represented by the product of appropriate elements of the local conditional distributions

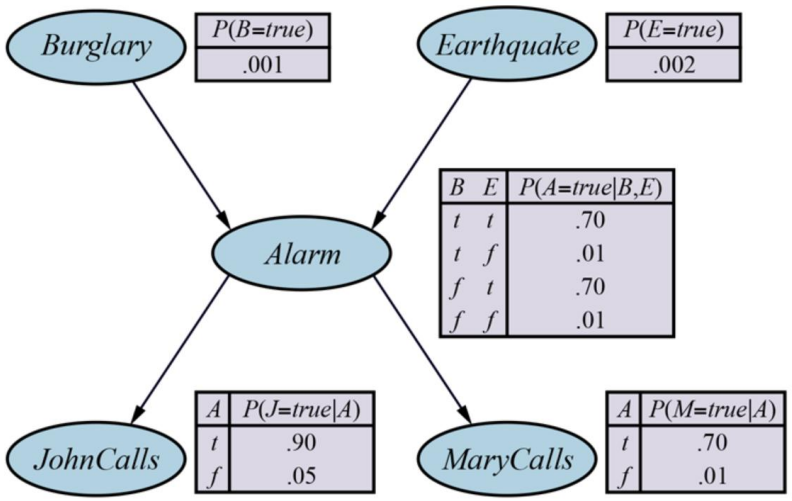


Fig. 13.2 R&N

- Consider: alarm sounds with neither burglary/earthquake and both neighbors call

$$P(j, m, a, \neg b, \neg e) = P(j|a)P(m|a)P(a|\neg b \wedge \neg e)P(\neg b)P(\neg e)$$
$$= .9 \times .7 \times .01 \times .999 \times .998 = .00628$$

Bayesian networks are frequently called Bayes nets

Bayes net semantics

- We can compute the marginal to answer just about any question related to this, e.g. John & Mary call when the alarm sounds and there is no burglary

$$P(j, m, a, \neg b) = \sum_{E \in e, \neg e} P(j|a)P(m|a)P(a|\neg b \wedge E)P(\neg b)P(E)$$

- Note that earthquake was not specified in the question; we computed the marginal probability to integrate/sum it out.

Constructing a Bayes network

- Nodes
  - Determine required random variables
  - Number them $X_1, X_2, \ldots, X_n$ (better if causes precede effects)
- Network edges

  for i = 1:n

  Find minimum parents(X_i) such that $P(X_i|X_{i-1}, \ldots, X_1) = P(X_i|parents(X_i))$

  Add edges $parents(X_i)$ to X_i

  Estimate conditional probability table $P(X_i|parents(X_i))$

Note: We are only concerned about direct influence, so Alarm influences MaryCalls, but Burglary and Earthquake do not.

Estimating the conditional probability table

- Estimated from training data
- Discrete – Use a frequentist model, e.g., $P(A = a|b) = \frac{count(a,b)}{count(b)}$
- Continuous - Fit distribution to data, e.g. $P(A = a|b) \sim n(\mu, \sigma^2)$, use the mean and variance of examples where B=b

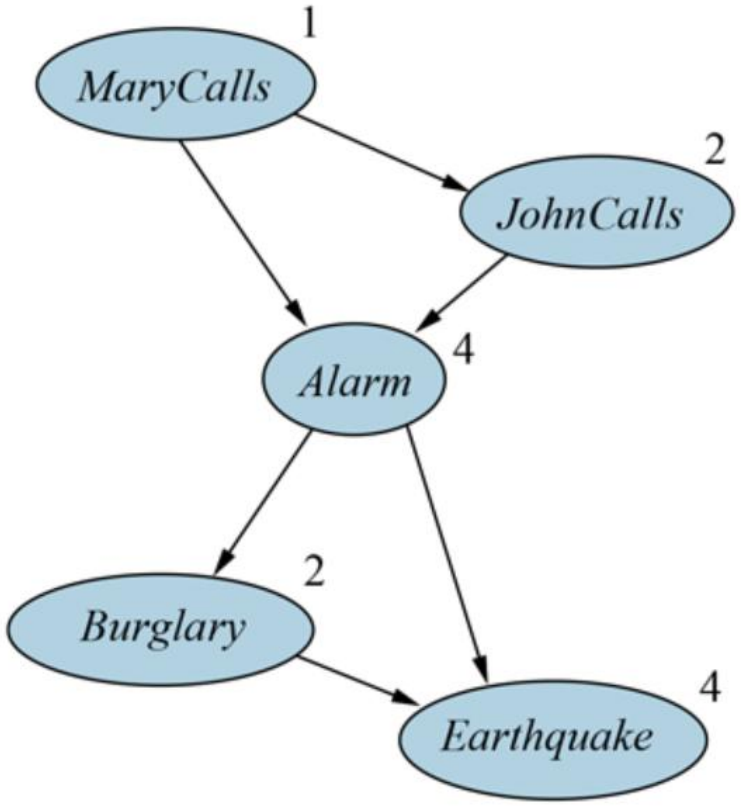
- Chapter 20 has more details on learning in probabilistic models. (See also any basic statistics book's chapter on maximum likelihood estimation)

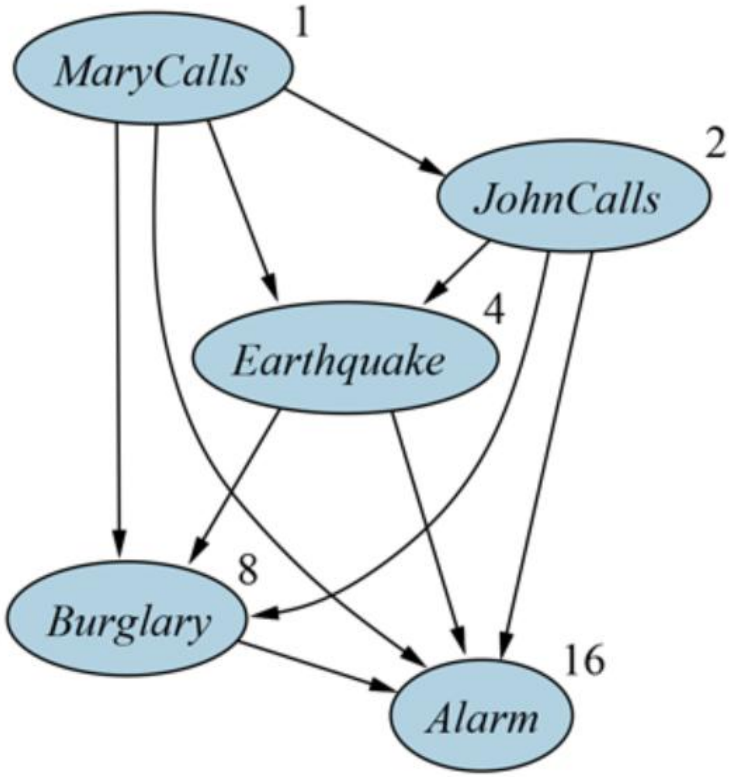
Variable order in Bayes network construction

- Construction depends on order
- Consider order: MaryCalls, JohnCalls, Alarm, Burglary, Earthquake
  - MC – no parents
  - JC – If MC, probably an earthquake, hence P(JC|MC)
  - A – Alarm more likely if both MC and JC call, therefore they are parents of A.
  - B – If we know alarm state, MC & JC do not give us additional information about whether this was a burglary or earthquake, hence P(B|A)
    [We assume that this is a minor earthquake, an example of laziness in uncertainty.]
  - E – If A, then it is more likely that there was an earthquake, but B would also cause an alarm and knowing this reduces the probability of E. Hence P(E|A,B)

Order of variables in construction matters

M,J,A,B,E





M,J,E,B,A

In general, better to order variables in what we think might be a causal manner. A poor ordering can **increase parameters**, forcing variables to take many parents, **enlarging CPTs exponentially** (e.g., binary node with k parents needs 2^k rows) and **make inference harder**

Bayesian nets with continuous variables

- Several options
  - Discretize – split into discrete values based on range
  - Use a parametric distribution, e.g., normal distribution
  - Non-parametric options possible, but beyond our scope

- Linear-Gaussian conditional distribution
  - Most common parametric distribution
  - Variance fixed, mean dependent on a continuous parent

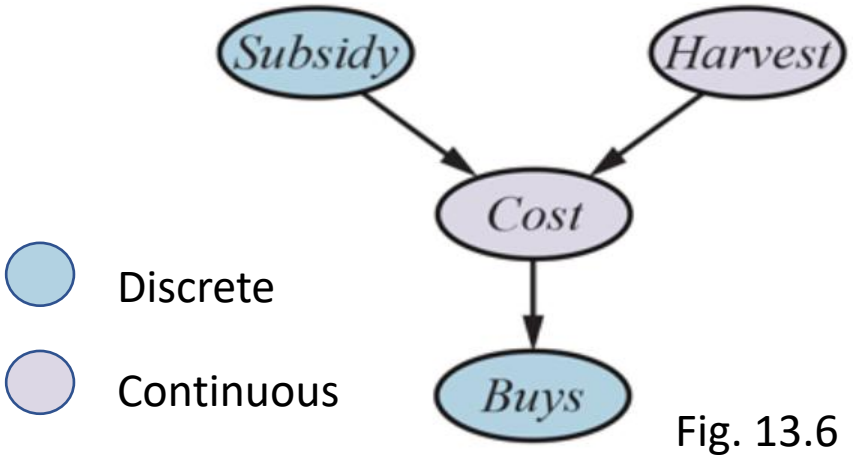
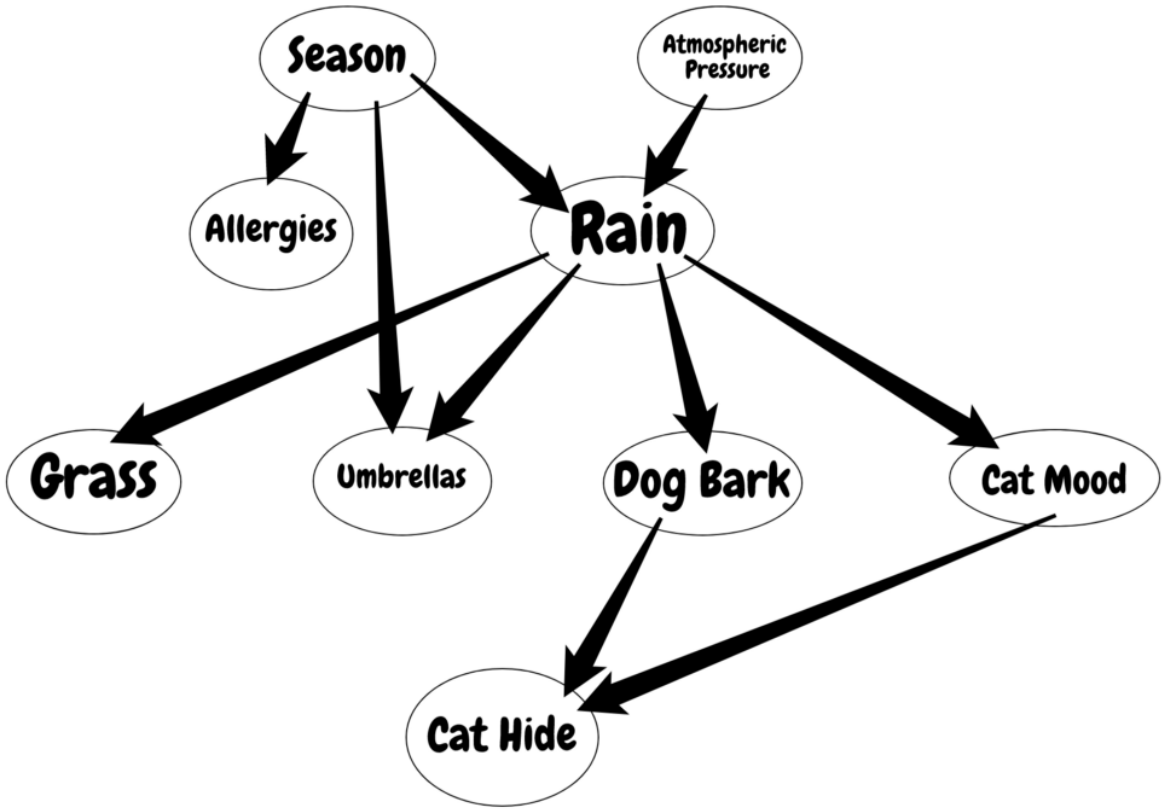


Fig. 13.6

Another example

We ask for the probability of “RAIN” if “SEASON” is “WINTER” and “DOG BARK” is “TRUE”.



Example by Atakan Güney, towardsdatascience.com

See 13.2.4 for a more complex case study

Applications:
Medical diagnosis,
fraud detection,
decision-making

Probabilistic Reasoning over Time - Markov models

- Often, we need to carry out probabilistic inference on temporal data or a sequence of observations where the order of observations matter.
- We require a model similar to a Bayesian Network, but one that grows over time to keep up with the latest evidences.
  - Bayesian Network: A general-purpose model capturing dependencies between any set of variables.
  - Markov Model: A time-driven model where the current state depends on the last.

    (Markov Property: the future depends only on the present, not the past)
    - A Hidden Markov Model is a special case of a Markov model in which the states are hidden, only outputs are visible.

Hidden Markov models

almostsuremath.com

- Used for modeling processes that have an unobservable state
- Example: magician is flipping
  - 2 coins behind a curtain with different odds of heads/tails
    Here one coin be fair, the other is biased
  - Magician has a process
    - Flip coin
    - Choose the next coin to flip

  - All we observe are sequences: H, H, H, H, T, H, T, T, H, …
    - What's the mostly likely sequence of coins used

- hidden Markov models let us model these types of systems,
- We don't know **which coin** he used at each flip — this is the **hidden** part

Markov property

- Observations (features)

$$O = \{o_1, o_2, \ldots, o_T\}$$

- Observations come from a discrete or continuous space and are independent of one another

$$o_i \in \Re^D \text{ or } o_i \in \{v_1, v_2, \ldots, v_{N_v}\}$$

- States $S = \{s_1, s_2, \ldots, s_{N_s}\}$

- State sequences[1]

$$q_1, q_2, \ldots, q_T \text{ where } q_i \in S$$

Markov property

- Let q_i be the state that we are in at time *i*. If $q_4 = fair$, we are using the fair coin for the 4th flip in our previous example.
- Chain rule (joint probabilities) states

$$\mathrm{P}(q_1, q_2, \dots, q_T) = \mathrm{P}(q_1) \prod_{i=2}^{T} \mathrm{P}(q_i | q_{i-1} q_{i-2} \cdots q_1)$$

- **Markov property specifies conditional independence after 1 step** (can be generalized to N steps)

$$\mathrm{P}(q_i | q_{i-1} q_{i-2} \cdots q_1) = P(q_i | q_{i-1})$$

Markov chains

- Sequence can be seen as moving from one state to another, dependent only upon the previous state:
  - Example, state: Pressure
  - P(high | yesterday high, day before changing) = P(high | yesterday high)

- barometer

Low pressure

Pressure changing

High pressure

Initial state distribution

- The Markov chain has a probability of starting in an initial state, denoted by the vector π

$$\pi = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} \begin{matrix} low \\ changing \\ high \end{matrix}$$

- In this example, the starting state has a uniform (equally probable) distribution.

State transition distribution

- Matrix A describes the state transition probabilities:

$$A = \begin{array}{c} \text{transition to} \\ \begin{array}{ccc} low & changing & high \end{array} \\ \begin{bmatrix} 1/2 & 1/4 & 1/4 \\ 1/4 & 1/2 & 1/4 \\ 1/4 & 1/4 & 1/2 \end{bmatrix} \end{array} \begin{array}{c} \text{transition from} \\ low \\ changing \\ high \end{array}$$

$$a_{ij} = \mathrm{P}(\, q_t = s_j | q_{t-1} = s_i)$$

Sometimes abbreviated as: $a_{ij} = \mathrm{P}(\, q_t = j | q_{t-1} = i)$

- P(high|low) = a_{13}=1/4

$$\sum_{j=1}^{N} a_{ij} = 1$$

State-dependent distributions

- For each state s, there is a probability of seeing an observation o: $b_s(o)$
- For our weather model example:

$$b_1(o) = \begin{cases} 3/4 & o = \text{rain} \\ 1/4 & o = \text{sun} \end{cases} \qquad b_2(o) = \begin{cases} 1/2 & o = \text{rain} \\ 1/2 & o = \text{sun} \end{cases} \qquad b_3(o) = \begin{cases} 1/4 & o = \text{rain} \\ 3/4 & o = \text{sun} \end{cases}$$

What are the odds of that?

$$\mathrm{P}(O_1 = rain, Q_1 = changing, O_2 = sun, Q_2 = high, O_3 = sun,\ \mathrm{Q}_3 = high)$$

which we abbreviate as: $\mathrm{P}(o_1, q_1, o_2, q_2, o_3, q_3)$

by chaining Bayes rule $\mathrm{P}(AB) = \mathrm{P}(A \mid B)\mathrm{P}(B)\ldots$

$$= \mathrm{P}(\mathrm{q}_1)\mathrm{P}(\mathrm{o}_1 \mid q_1)\mathrm{P}(q_2 \mid q_1 o_1)\mathrm{P}(o_2 \mid q_2 q_1 o_1)\mathrm{P}(q_3 \mid o_2 q_2 q_1 o_1)\mathrm{P}(o_3 \mid q_3 o_2 q_2 o_1 q_1)$$

As o_i only dependent on state q_i

$$= \mathrm{P}(\mathrm{q}_1)\mathrm{P}(\mathrm{o}_1 \mid q_1)\mathrm{P}(q_2 \mid q_1 o_1)\mathrm{P}(o_2 \mid q_2)\mathrm{P}(q_3 \mid o_2 q_2 q_1 o_1)\mathrm{P}(o_3 \mid q_3)$$

and as we have a first order Markov chain

$$= \mathrm{P}(\mathrm{q}_1)\mathrm{P}(\mathrm{o}_1 \mid q_1)\mathrm{P}(q_2 \mid q_1)\mathrm{P}(o_2 \mid q_2)\mathrm{P}(q_3 \mid q_2)\mathrm{P}(o_3 \mid q_3)$$

$$= \pi_{changing} b_{changing}(rain) a_{changing,high} b_{high}(sun) a_{high,high} b_{high}(sun)$$

or using our state numbers: $\pi_2 b_2(rain) a_{2,3} b_3(sun) a_{3,3} b_3(sun)$

s_1
Low pressure

s_2
Pressure changing

s_3
High pressure

The *hidden* in hidden Markov model

- Barometer let us observe the state.

- Suppose we cannot observe state.

- Many state sequences are possible,
  each sequence has a probability of occurrence.

Hidden Markov models (HMM)

- Let $\Phi(A,B,\pi)$ denote a HMM where:
  - A – NxN state transition matrix. a_{ij} denotes the probability of transitioning from state i to j.
  - B – $\{b_j(k)\}$ – Set of state-dependent probability distributions. $1 \leq j \leq N_s$, k in O
  - π – Initial state distribution. π_j is the probability of starting in state j. $1 \leq j \leq N_s$

Hidden Markov models

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- Example:
  - You suspect the magician has two coins:
    - Coin A: fair coin (P(H) = 0.5, P(T) = 0.5)
    - Coin B: biased coin (P(H) = 0.8, P(T) = 0.2)

  - But you don’t know **which coin** he used at each flip — this is the **hidden** part of the model.

Components of Hidden Markov models

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- Hidden States: These are the things you can't directly observe.
  - State 1: Coin A
  - State 2: Coin B

- Observations: These are what you actually observe.
  - Observations: "Heads" or "Tails“ or sequene “Heads, Heads, Tails”

Components of Hidden Markov models

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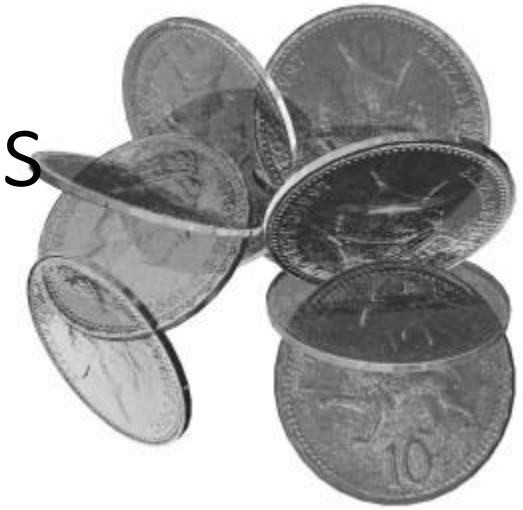
- Initial Probabilities: Probability of starting with Coin A or B.
  - π – Initial state distribution. π_j is the probability of starting in state j
  - π_1 = P(start with A) = 0.6
  - π_2 = P(start with B) = 0.4

- Emission Probabilities: Probability of observing a certain result given a state.
  - If using Coin A: fair coin ($b_1(H) = 0.5$, $b_1(T) = 0.5$)
  - If using Coin B: biased coin ($b_2(H) = 0.8$, $b_2(T) = 0.2$)
  - B – $\{b_j(k)\}$ – Set of state-dependent probability distributions

Components of Hidden Markov models



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- Transition Probabilities: Probability of switching from one state (coin) to another.
  - $a_{11} = P(A \rightarrow A) = 0.9$,
  - $a_{12} = P(A \rightarrow B) = 0.1$
  - $a_{22} = P(B \rightarrow B) = 0.8$,
  - $a_{21} = P(B \rightarrow A) = 0.2$
  - A – NxN state transition matrix, a_{ij} denotes the probability of transitioning from state i to j

(In this case, the magician tends to stick with the same coin)

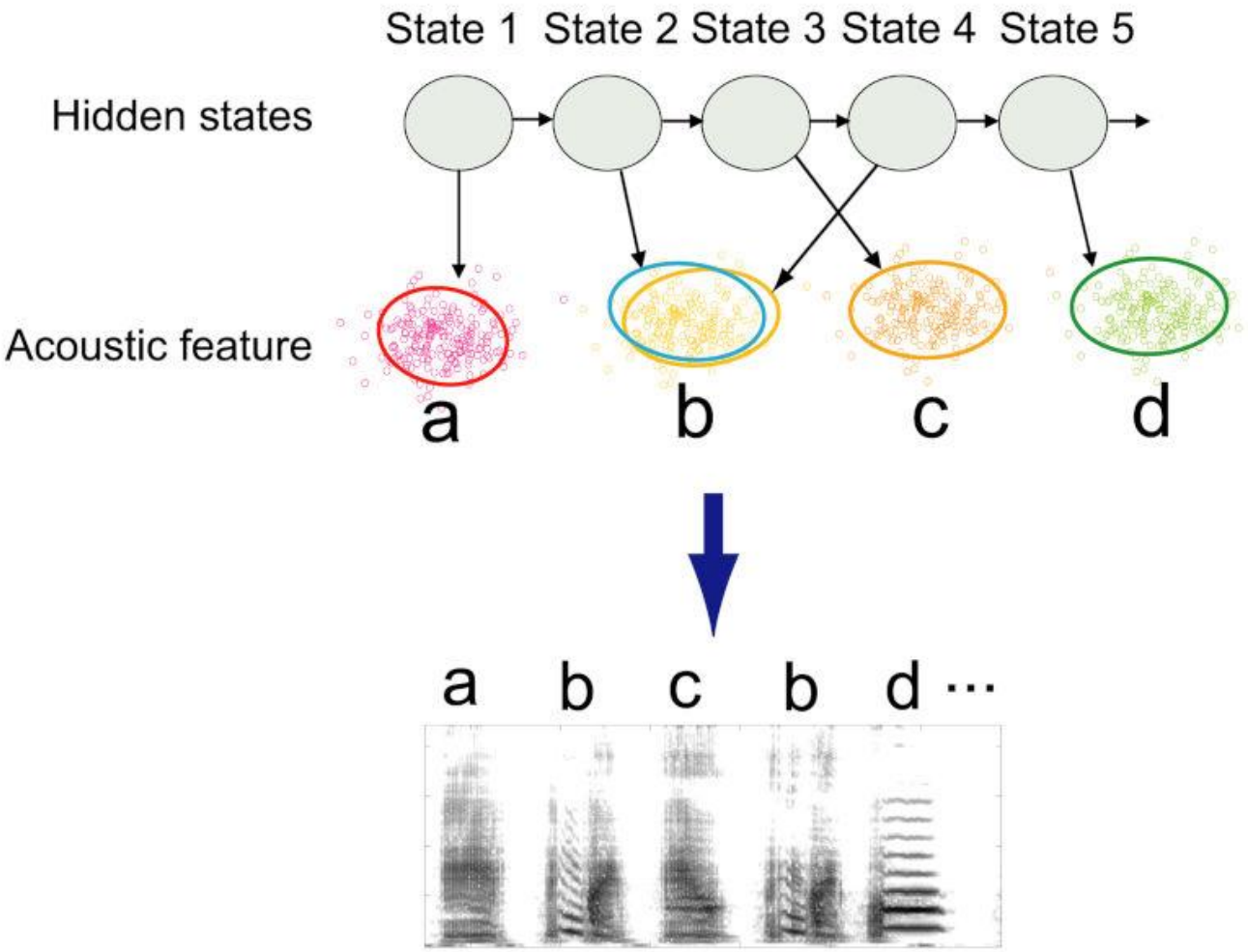
Top 3 List for HMMs

1. What is the probability of a given sequence O given model Φ?
2. What state sequence is most likely to account for O in model Φ?
3. How can we improve the parameters of Φ to better account for O?

Top 3 Applications of HMMs

1. Prediction: Estimate which coin was likely used at each step. (e.g., Viterbi algorithm)
2. Likelihood: Compute how likely a particular sequence is given the model. (e.g., Forward algorithm)
3. Learning: Adjust the model based on observed data (Baum-Welch algorithm)

Isolated Word Recognizer



Each word is modeled by its own HMM

- Hidden states ≈ phonemes or subword units
- Observations = acoutic feature vectors

Parameters:

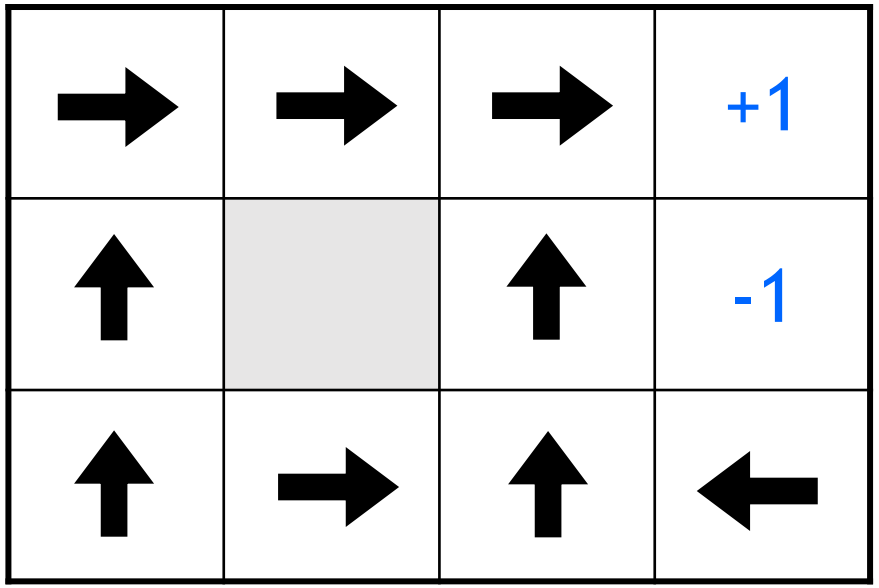
- Transition probabilities: likelihood of moving between states.
- Emission probabilities: likelihood of observing features given a state.

Word Recognition

- Compute the likelihood of the observation sequence under each word model using the Forward algorithm.
- Choose the word whose HMM gives the highest probability.

Reinforcement Learning – Learning a policy

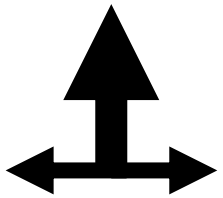
Reward for each step: -0.1



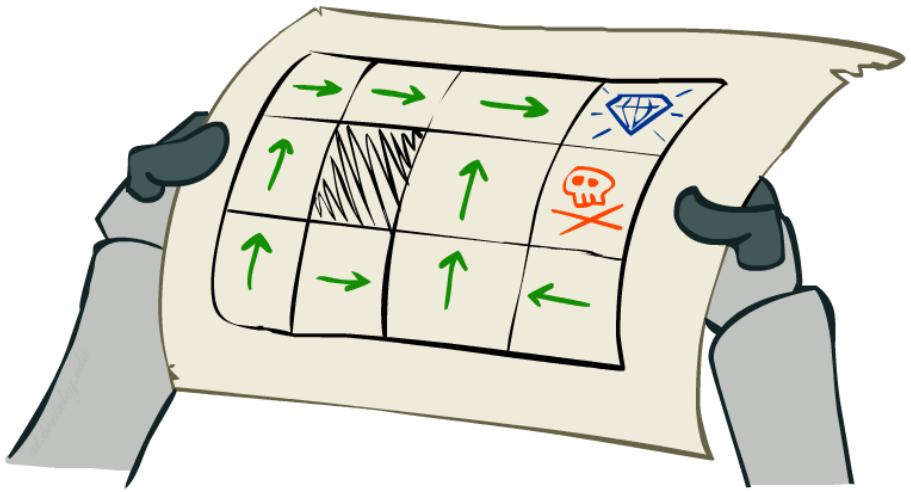
actions: UP, DOWN, LEFT, RIGHT

UP

80% move UP
10% move LEFT
10% move RIGHT



reward +1 at [4,3], -1 at [4,2]
reward -0.1 for each step

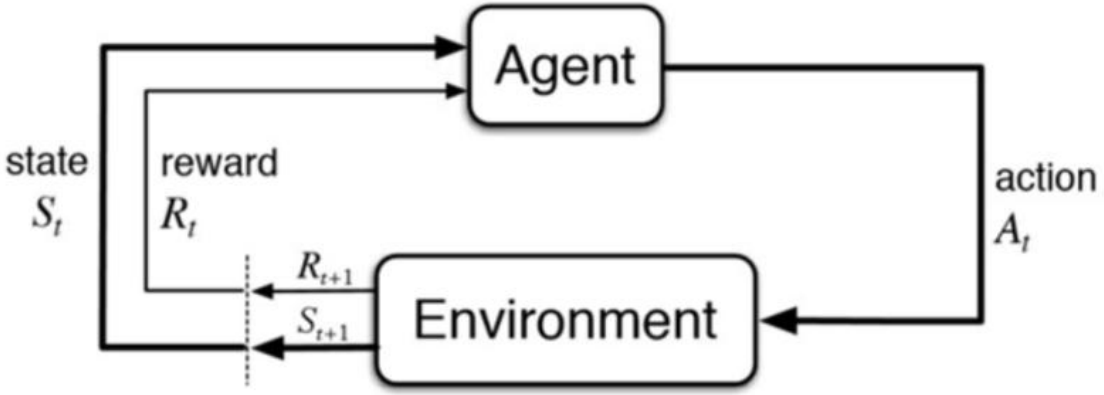


Reinforcement Learning - Markov Decision Process (MDP)

- Learning a Policy

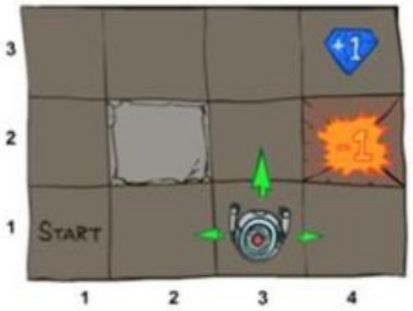
An MDP is defined by:

- Set of states S.
- Set of actions A.
- Transition function $P(s'|s,a)$.
- Reward function $r(s,a,s')$.
- Start state s_0.
- Discount factor γ.
- Horizon H.

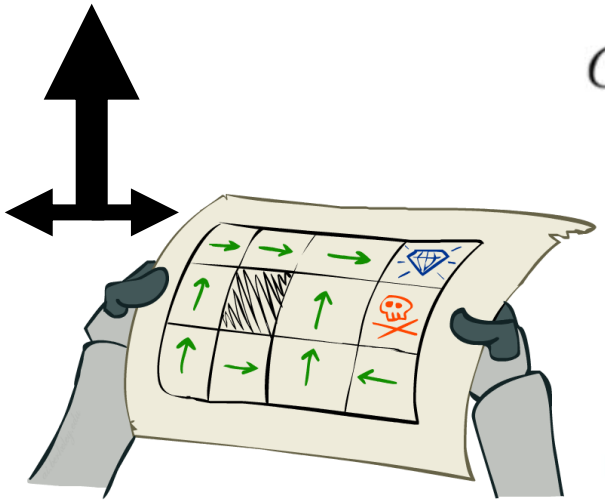


Return:

$$G_t = R_{t+1} + \gamma R_{t+2} + \ldots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$



π:



Policy: $\pi(a|s) = \Pr(A_t = a | S_t = s) \quad \forall t$

Goal: $\arg\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{H} \gamma^t R_t | \pi\right]$